CODERED '26 | INDUSTRY CHALLENGE BRIEF

# Problem Statement 5
# One-Token Video

Compute-efficient, high-quality AI video generation and editing

**OBJECTIVE** Overcome the high computational cost and token consumption of current AI video generation tools by engineering an efficient video editing and generation platform that produces high-quality clips while consuming minimal processing tokens.

## 1. Challenge Overview

**Company / Partner Name:** MyStartupWave

**Challenge Title:** One-Token Video

**Industry / Track:** Generative AI / Media & Content Technology

**Challenge Contact:** MyStartupWave Challenge Desk (mystartupwave@gmail.com)

**Challenge Type:** Product Innovation

**Brief Summary:** Current AI video generation tools are expensive and slow because they consume large amounts of compute and tokens per second of output, putting them out of reach for most creators and small businesses. Teams will engineer an efficient video editing and generation platform that produces high-quality short clips while consuming dramatically less compute, and will prove the quality-versus-cost trade-off with transparent measurements.

## 2. Industry Context

Short-form video dominates digital marketing, social media and education, and demand for video content far outstrips the capacity of creators and small businesses to produce it. Text-to-video and image-to-video models have improved dramatically, but generating even a few seconds of video can require significant GPU time and costly credit-based pricing, with long waits and frequent regeneration when results are wrong.

Because most edits mean regenerating from scratch, AI video remains impractical for iterative creative work and for price-sensitive markets such as India. Techniques such as keyframe reuse, latent caching, efficient interpolation, model distillation, smart composition of generated and existing assets, and careful prompt planning offer a route to much cheaper generation, if they are combined into a product that keeps quality high.

## 3. Problem Statement

- **Problem and who is affected:** Content creators, marketers, small businesses, educators and startups need frequent short-form video but cannot afford high per-clip generation costs or long processing times.
- **When and where it occurs:** Whenever a video must be produced or revised quickly (social campaigns, product promos, explainers, ads), and especially during iterative editing, where small changes trigger full, expensive regenerations.
- **Limitations of current solutions:** Leading tools charge per second of generated video and regenerate entire clips for minor edits; capable open-source models need high-end GPUs; outputs often suffer from temporal flicker and inconsistent subjects; and users have little visibility into, or control over, compute spend.

- **Why it is worth solving:** Credit-based pricing and long render queues are common complaints among AI video users; demand for short-form video keeps growing; and strong research momentum on efficient video diffusion, distillation and caching indicates that large efficiency gains are achievable.

## 4. Target Users and Stakeholders

**Primary users:** Independent creators, social media managers, and small and medium businesses producing short-form video.

**Secondary users / stakeholders:** Marketing agencies, ed-tech content teams, and developers who integrate video generation into their own products.

**Current user journey or workflow:** Write a prompt → wait for generation → receive a clip that is partly wrong → regenerate the whole clip, spending credits again → export → edit in a separate editor → repeat for every variant or aspect ratio.

**Key user pain points:** High cost per usable clip; slow turnaround; full regeneration for small changes; inconsistent characters and objects across shots; no visibility into compute spend; and separate tools for generation and editing.

## 5. Product Opportunity

- **What users should accomplish:** Generate and edit short, high-quality video clips from text, images or existing footage at a fraction of today's compute cost.
- **Meaningful improvement:** Incremental edits that change only what needs changing (one shot, an object, text, style or duration) without regenerating the whole video.
- **What makes it different:** A transparent cost meter that shows users exactly how much compute each output consumes, backed by a measured, reproducible quality-versus-cost comparison.

## 6. Expected Product Capabilities

**Must-have capabilities:**

- Text-to-video and image-to-video generation of short clips (at least 10–15 seconds, 720p or higher, 24 fps) with coherent motion and consistent subjects.
- An efficiency architecture that substantially reduces compute or tokens per second of output compared with a naïve baseline, e.g., keyframe generation with interpolation, latent/feature caching, reuse of shared scenes, low-resolution drafts with selective upscaling, or composition of generated and existing assets.
- Incremental editing: modify a single shot, segment or element and regenerate only the affected portion.
- A built-in timeline editor supporting trimming, sequencing, text overlays, transitions and an audio track.
- A per-job cost and performance meter showing GPU-seconds or tokens consumed, generation time and estimated cost, compared against the team's own baseline pipeline.
- Quality evaluation using objective metrics (e.g., CLIP-based text–video alignment, temporal-consistency/flicker score, FVD where feasible) alongside human ratings.

**Nice-to-have capabilities:**

- A draft/preview mode that produces a low-cost storyboard or animatic before committing to a full render.
- Character and brand consistency across multiple clips (reference-image locking).
- Automatic aspect-ratio adaptation (16:9, 9:16, 1:1) without full regeneration.
- AI voice-over, captions and music synchronisation.
- Watermarking or C2PA labelling of AI-generated video.

**Expected user interactions:** The user writes a prompt or uploads images or footage → reviews a low-cost storyboard → generates the clip → edits individual shots on the timeline while seeing the cost of each action → exports in the required format.

**Expected demonstration / deliverable:** Live generation of at least two distinct clips from prompts and one single-shot edit; a side-by-side comparison against a baseline pipeline on identical prompts, showing compute consumed, time taken and quality metrics; and the final exported videos.

## 7. Technical and Operational Constraints

**Technical or platform constraints:** Any model stack may be used (open-source preferred). Teams must define and implement a fair baseline (e.g., the same base model generating full clips directly) to measure efficiency gains against. The measurement methodology must be documented and reproducible.

**Performance, accuracy, or latency requirements:** At least a 5× reduction in compute (GPU-seconds or tokens) per second of output video versus the team's baseline, with no more than a 10% drop on the chosen quality metrics and comparable human-rated quality. A single-shot edit must cost less than 25% of a full regeneration. A 10-second 720p clip should generate in under 3 minutes on a single consumer or free-tier GPU (e.g., T4 or A10 class).

**Security, privacy, or compliance requirements:** No generation of deepfakes of real people without consent, no sexual or violent content, and respect for copyrighted characters and brands. Uploaded media must be used only for the user's own job.

**Hardware, connectivity, or deployment limitations:** Must be runnable on a single GPU available through free or low-cost tiers (e.g., Google Colab, Kaggle) or through APIs. The front end must run in a browser.

**Budget or resource constraints:** Total compute used during the event should stay within free tiers or modest credits. Teams must report cost per second of output video.

**Domain-specific safety requirements:** Content-safety filtering on prompts and outputs, and clear labelling of generated videos as AI-generated.

## 8. Available Resources and Support

Resources marked "Confirmed" are guaranteed by the partner. "Permitted" items may be used by teams at their own discretion and cost. "Under consideration" items are not yet confirmed.

**APIs, SDKs, developer tools, or technology credits:** Confirmed: none provided by the partner. Permitted: open-source video models (e.g., Stable Video Diffusion, CogVideoX, AnimateDiff, LTX-Video, Wan), video tooling such as FFmpeg, or generative AI APIs available to the team.

**Sample datasets or permitted data sources:** Openly licensed stock footage and images, team-created media, and public video-generation benchmark prompts, used within their licences.

**Sandbox environments or test infrastructure:** Not provided. Teams use free GPU tiers (e.g., Google Colab, Kaggle) or their own resources.

**Documentation and reference materials:** Model cards and papers for the chosen video models; FFmpeg documentation; literature on efficient diffusion (distillation, caching, frame interpolation).

**Mentors, domain experts, or technical sessions:** Confirmed: MyStartupWave engineers during the scheduled mentoring rounds (subject to the event schedule).

**Other resources or support:** Under consideration: a technical session on efficient diffusion inference.

## 9. Business and Adoption Context

**Intended customer or beneficiary:** Creators, SMB marketers and agencies (SaaS); developers (API).

**Current alternatives or competing solutions:** Commercial AI video generators (e.g., Runway, Pika, Luma, Kling), template-based editors (e.g., CapCut, Canva) and self-hosted open-source models.

**Potential value created:** An order-of-magnitude lower cost per usable clip, faster iteration, fewer wasted regenerations, and AI video that is affordable in price-sensitive markets.

**Potential adoption pathway:** A freemium creator tool with pay-as-you-go credits; an API for platforms that embed video generation; agency plans.

**Commercial or operational considerations:** GPU cost and availability, model licensing terms (some open models restrict commercial use) and content-moderation obligations.

**Known assumptions or uncertainties:** Efficiency techniques are assumed to generalise across prompt types, and users are assumed to accept slight quality trade-offs in exchange for large cost savings.

## 10. Success Indicators

**Functional outcomes:** Generation, incremental editing, timeline editing, cost metering and export working end to end.

**User or customer outcomes:** Users can produce a publishable 15-second clip, including at least one revision, in under 10 minutes; human raters judge quality comparable to the baseline.

**Technical performance indicators:** ≥ 5× compute reduction per output second versus baseline with ≤ 10% quality-metric drop; single-shot edits ≤ 25% of full-regeneration cost; a 10-second 720p clip in under 3 minutes on a single T4/A10-class GPU.

**Business or operational indicators:** A credible cost-per-clip and pricing model showing affordability for SMB creators.

**Safety, reliability, or compliance indicators:** Prompt and output safety filters active; generated videos labelled; no misuse of real people’s likenesses in demos.

## 11. Scope and Boundaries

**In scope:** An efficient generation pipeline, incremental editing, a timeline editor, a cost meter, quality evaluation and the baseline comparison.

**Out of scope:** Training a foundation video model from scratch, long-form video (over 60 seconds), real-time or live video generation, and 4K output.

**Permitted assumptions or simplifications:** Teams may build on existing pre-trained models and fine-tune or distil them. Short clips (10–30 seconds) are sufficient. Quality metrics may be computed on a small benchmark prompt set (e.g., 10–20 prompts).

**Dependencies (systems, approvals, resources):** GPU access and pre-trained model weights with compatible licences.

## 12. Industry Engagement

**Will the company provide mentors or subject-matter experts?** Yes. Engineers from MyStartupWave will be available during the scheduled mentoring rounds for architecture, product and feasibility reviews.

**Will teams receive feedback during development?** Yes. Teams will receive structured feedback at the designated mid-event checkpoint(s), focused on scope, technical approach and progress against the success indicators.

**Will the company participate in evaluation or demonstrations?** Yes. MyStartupWave representatives will attend the final demonstrations and take part in evaluation alongside the CODERED ’26 jury.

**Is further validation, pilot discussion, or product exploration possible after the event?** Possible. Teams with exceptional solutions may be invited to a follow-up discussion on further validation, mentoring or pilot exploration.

Any follow-on engagement is subject to mutual interest and separate agreement.

## 13. Intellectual Property and Participation Terms

The following terms apply to all participating teams and are communicated before development begins. Participation does not imply any automatic transfer of ownership or commercial rights; any subsequent arrangement must be agreed separately.

**Data and resource usage conditions:** Use only openly licensed or team-created media and models whose licences permit the intended use. Do not upload or generate content depicting real people without their consent.

**Confidentiality requirements, if any:** None. This brief and any materials shared for this challenge are non-confidential.

**Intellectual property and ownership terms:** Teams retain full ownership of the code, designs, models and other intellectual property they create. Participation does not transfer any ownership or commercial rights to MyStartupWave.

**Restrictions on using company names, data, or materials:** Do not use MyStartupWave's name or branding without written permission. Commercial AI video products may be referenced only descriptively, for comparison.

**Terms governing subsequent pilots, licensing, investment, employment, or commercial collaboration:** Any pilot, licensing, investment, employment or commercial collaboration after the event is subject to mutual interest and a separate written agreement.

## 14. Partner Contact and Clarifications

**Primary contact:** MyStartupWave Challenge Desk

**Email:** mystartupwave@gmail.com

**Reference documentation / resource links:** This challenge brief; model cards of the chosen video models; FFmpeg documentation.

**Clarification process:** Send questions in writing to the challenge contact or raise them through the official CODERED '26 helpdesk. Answers that affect scope or evaluation will be shared with all teams on this challenge at the same time, to keep the competition fair.

## Submission Checklist

- ☑ The problem is real, relevant, and clearly stated.
- ☑ Target users and desired outcomes are identified.
- ☑ Teams have freedom to develop their own solutions.
- ☑ A working prototype is reasonably achievable within the challenge timeline.
- ☑ Business value and adoption can be evaluated.
- ☑ Technical constraints and available resources are stated.
- ☑ Data, safety, confidentiality, and IP terms are clear.
- ☑ Evaluation expectations are communicated consistently.

**THE GUIDING PRINCIPLE**

**The company defines the opportunity. The team defines the solution.**